Data Policies
A Sustainability Checklist

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Sustainability encompasses the three E's of Environment, Equity, and Economics(*)

Environment

- [ ] Tracking energy consumption of the data infrastructure
- [ ] Promoting Green IT practices (hardware/software)
- [ ] Investing in resource-efficient data infrastructures

Equity

- [ ] Lowering the technological threshold to data access (low-bandwidth environments)
- [ ] Increasing data findability for non-scientific stakeholders
- [ ] Complying with IPR and ensuring recognition is given even when data have no IPR

Economics

- [ ] Monitoring costs for data storage, management, and curation
- [ ] Exploiting economies of scale in handling data
- [ ] Understanding if and how current data can help pay for future data storage

Data Policies are official documents created by organisations to provide guidelines for the management of the research data they generate or host on behalf of third parties. Some data policies aim at a sustainable data management, but what does this mean in practice?

The checklist considers the three areas that are routinely included in sustainability assessments, i.e., environmental, social, and economic, and lists the main points that a data policy should address if sustainable data management is an aim of the organisation.

From an **environmental perspective**, it is essential that an organisation understands how energy-intensive its data infrastructure is. Ideally, organisations running their own data centres should also account for a larger set of resources used (e.g., water). Only by identifying critical aspects in the usage of energy and natural resources, an organisation can lower its environmental impact. Green IT and green software, i.e., selecting hardware that consumes less electricity, writing optimised code that requires less computational resources, opting for data formats that need less storage space, and similar strategies, are a starting point in making environmentally-conscious choices. Data policies should also describe how an organisation plans to invest in increasing the resource-efficiency of its data infrastructures to guarantee that they keep pace with the manifold increase in the volume of research data without multiplying their environmental impact.

From a **social perspective**, scientific organisations should not limit themselves to mention open science and the FAIR principles in their data policies. They should rather address how these high-level principles and practices find practical application in their infrastructures for research data. "Open data" is an enticing term, but making data available free of charge is not enough, if data access is then limited to those who have high-bandwidth. In addition, the availability of research data for a wider public of stakeholders requires to rethink the findability aids currently used. Metadata standards are typically developed by scientists for scientists and may be difficult to navigate for policy makers, governmental advisors, etc. Last, but not least, not all research data qualify for IPR protection. Therefore, data policies should lay out clear and actionable processes to identify whether there is IPR on research data, who is the IPR holder, and what is the best way to license such data. If there is no IPR and data cannot be licensed, data collectors should still receive proper credit for their work, and requirements and permissions in using the data should be clearly communicated to the users.

From an **economic perspective**, data policies should give an insight into the the processes adopted by research organisations to monitor costs for data storage, management, and curation. Given the staggering increase in research data and the recommended archival time of ten years or more, the financial stability of research data management services requires a careful strategy. Drafting or revising a data policy may provide the best opportunity to discuss how an organisation intends to use economies of scale (in the cloud, in federated research infrastructures, etc.) to stabilise its financial commitments for data storage and management. It is also an opportunity to discuss how the open-knowledge commitment of publicly-funded organisations can be combined with market and non-market mechanisms, such as fee-based models, that may support research data infrastructures in the long term.

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(*) **R. Brinkmann, "Defining Sustainability". In: *The Palgrave Handbook of Global Sustainability*. https://doi.org/10.1007/978-3-030-38948-2_1-1**

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